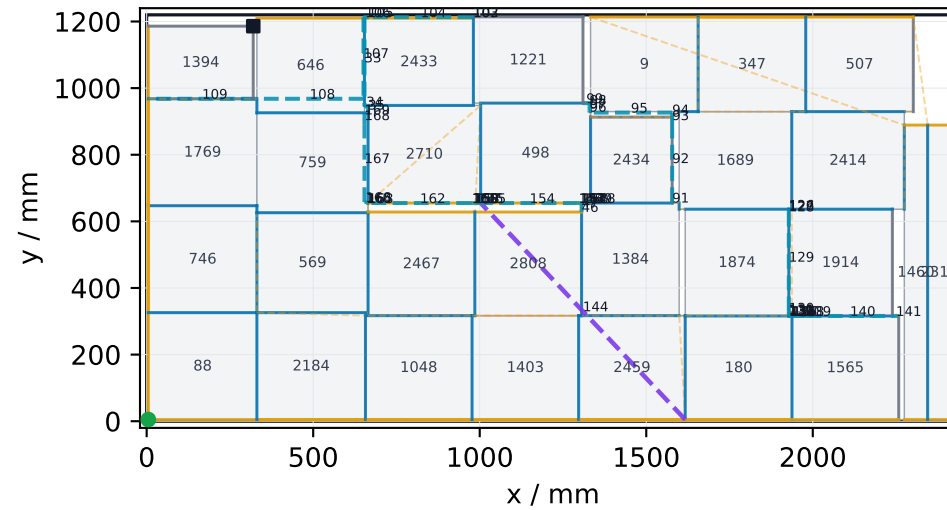


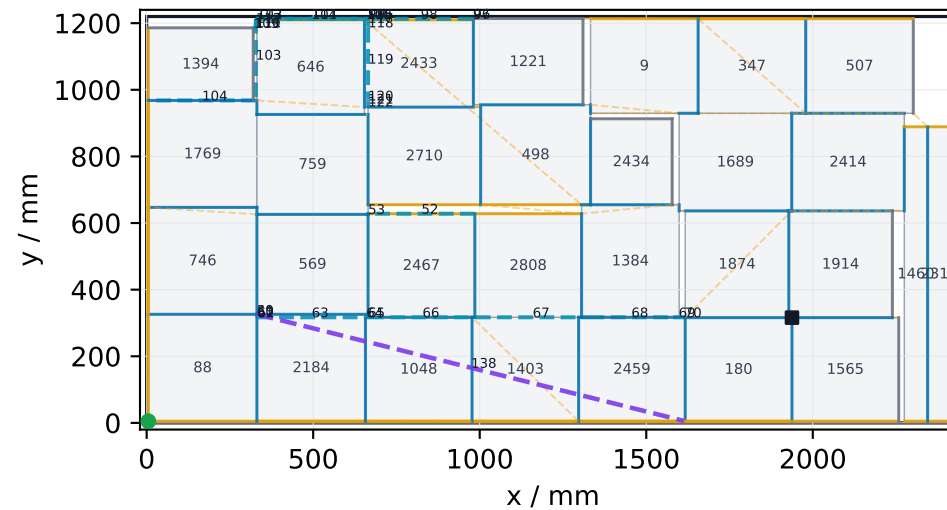
Route comparison: rectangles=30, panel=2440 x 1220 mm

— shared-edge cut — collinear chain - - travel - - safe lift × stability event
— near-shared cut — single edge - - detour - - penalized travel

Process-aware: runtime=212.5 ms, cost=13483.9, machining=38095.4, stability=0, safe_lift=1, detour=61



Process-aware beam: runtime=507.4 ms, cost=12563.1, machining=37174.6, stability=0, safe_lift=1, detour=37



Event-gated adaptive beam+LS: runtime=995.1 ms, cost=10324.3, machining=34935.8, stability=0, safe_lift=1, detour=10

